<INTRODUCTION TO DATA MANAGEMENT>

PROJECT REPORT

(Project Semester January-April 2025)

(Indian Agriculture Analysis Dashboard)

Submitted by

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Section KM007

Course Code INT217

Under the Guidance of

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Discipline of CSE/IT

Lovely School of Computer Science & Engineering

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that Shabad Grover bearing Registration no.12301468 has completed .INT217 project titled, **"*City Cycling Trends*"** under my guidance and supervision of Maneet Kaur. To the best of my knowledge, the present work is the result of his original development, effort and study.

Name of the Supervisor – Maneet Kaur
Designation of the Supervisor- Assistant Professor
School of Computer science & Engineering
Lovely Professional University
Phagwara, Punjab.

Date: 12 April 2025

DECLARATION

I, Shabad Grove, student of Computer science & engineering under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 12 April 2025 Name of the student

Registration No. 12301468 Shabad Grover

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my mentor and guide Maneet Kaur for their valuable guidance and support throughout this project. I am also thankful to Lovely Professional University for providing me with the platform and resources needed to complete this minor project. Lastly, I thank my peers and family for their encouragement.

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1. Introduction

Bike-sharing systems have become a key component of urban transportation, offering a sustainable and efficient mobility solution. However, understanding usage patterns, demand fluctuations, and external influences like weather or time-of-day is critical for optimizing operations. This project focuses on building an interactive Excel dashboard that visualizes and analyzes bike-sharing usage data. The dashboard reveals patterns in ride counts, peak usage times, and the impact of environmental factors.

2. Source of Dataset

The dataset used for this project is sourced from the publicly available bike-sharing dataset. It comprises hourly records detailing ride counts, weather conditions, time indicators (such as hour, day, month, and year), and other relevant fields. The dataset contains over 17,000 entries, enabling a comprehensive analysis of user behavior and operational patterns across different periods.

3. Dataset Preprocessing

Before analysis, the dataset underwent a series of cleaning and formatting steps to ensure accuracy and usability:

- **Irrelevant Columns Removed:** Only key columns such as date, season, year, month, hour, holiday, weekday, workingday, weather, temp, and cnt (total ride counts) were retained.
- **Data Cleaning:** Duplicate values were eliminated and missing or erroneous entries were checked and addressed.
- **Table Conversion:** The cleaned dataset was converted into an Excel Table to facilitate dynamic referencing and improve interactivity.
- **Pivot Table Setup:** PivotTables were created to derive metrics like total rides, average rides per day, and hourly trends, forming the backbone of the dashboard.

4. Analysis on Dataset

i. General Description

The analysis was conducted using Microsoft Excel. The dashboard incorporates PivotTables, slicers, and a variety of charts (line, bar, pie/doughnut, and heatmaps) to illustrate different aspects of the data.

ii. Key Insights & Visualizations:

1. **Dashboard Overview:**

- **KPIs:** Display Total Rides, Average Rides per Day, and the Most Active Hour.
- **Interactive Slicers:** Allow filtering by year, month, weekday, and weather for dynamic insights.

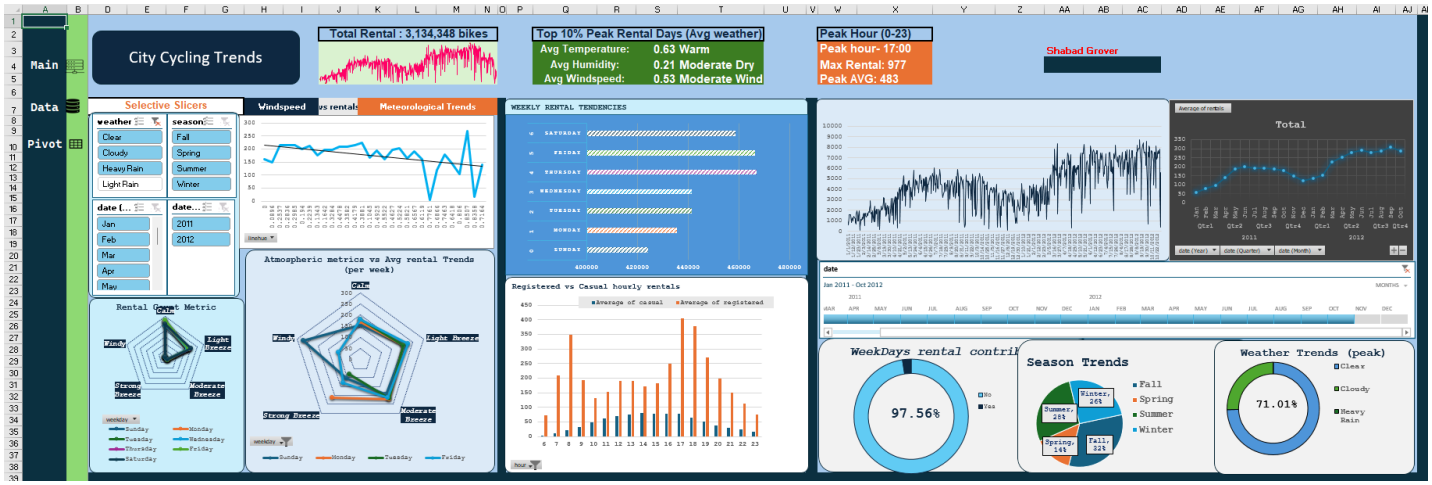
2. **Trend Analysis:**
   - **Monthly Trend (Line Chart):** Shows the progression and seasonality of bike usage over time.
   - **Weekday Analysis (3D Bar Chart):** Highlights the variation of rides across different days of the week.
3. **Category Distribution:**
   - **Ride Distribution by Weather (Pie Chart):** Visualizes the proportion of rides occurring under different weather conditions.
   - **Ride Distribution by Season (Doughnut Chart):** Illustrates how seasonal changes affect bike usage.
4. **Heatmap:**
   - Conditional formatting is applied to a pivot table grid to highlight peak ride hours and days visually.

iii. Visual Enhancements

All charts were enhanced using Excel's formatting tools—slicers for interactivity, custom color schemes for clarity, and conditional formatting for additional visual cues.

8. Linked In post - https://www.linkedin.com/feed/update/urn:li:activity:7316786536094568450/

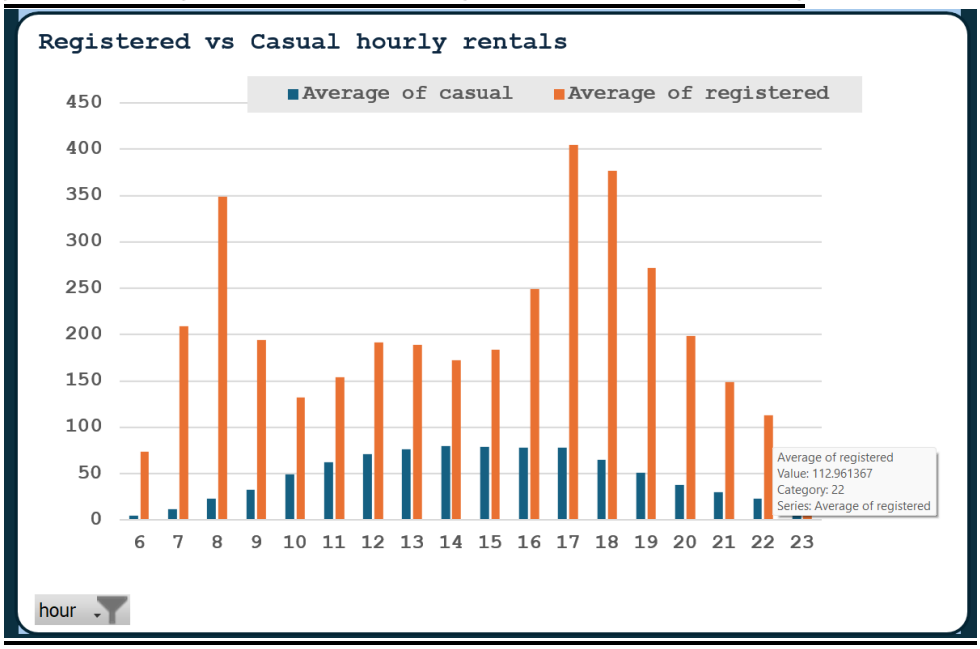
Dashboard


Main
Data
Pivot
City Cycling Trends
Total Rental : 3,134,348 bikes
Top 10% Peak Rental Days (Avg weather)
Avg Temperature: 0.63 Warm
Avg Humidity: 0.21 Moderate Dry
Avg Windspeed: 0.53 Moderate Wind
Peak Hour (0-23)
Peak hour- 17:00
Max Rental: 977
Peak AVG: 483
Shabad Grover
Selective Slicers
Windspeed vs rentals
Meteorological Trends
WEEKLY RENTAL TENDENCIES
Total
Atmospheric metrics vs Avg rental Trends (per week)
Registered vs Casual hourly rentals
WeekDays rental contrib
97.56%
Season Trends
Weather Trends (peak)
71.01%

OBJECTIVES

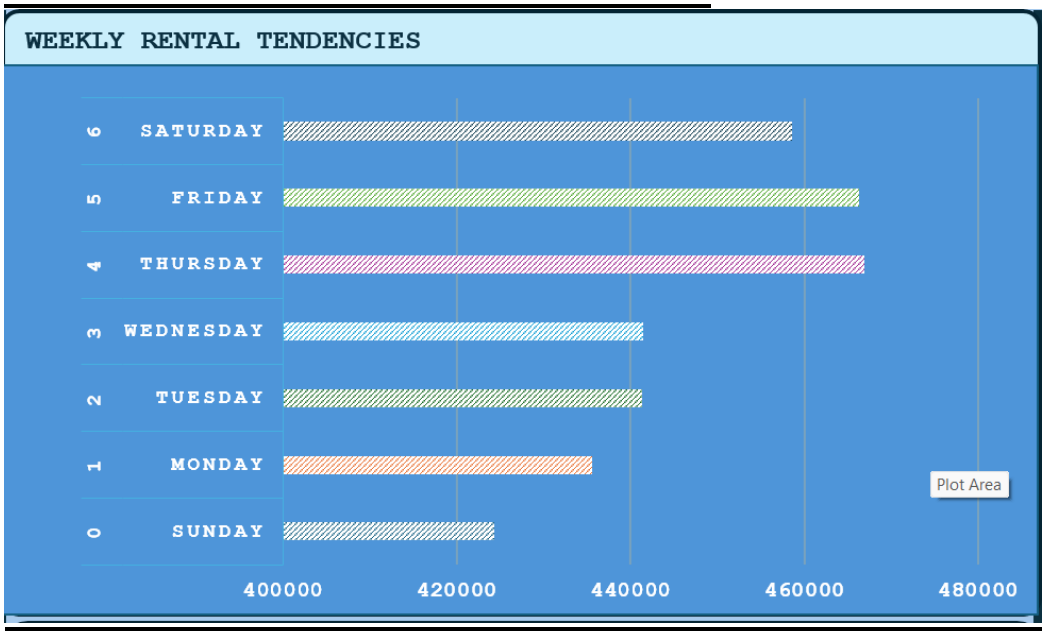
1) Registered vs Casual hourly rentals.

Row Lab	Average of casual	Average of registered
6	4.32	73.37
7	11.63	209.06
8	22.75	348.85
9	32.46	194.10
10	49.14	132.07
11	62.61	154.00
12	71.09	191.71
13	76.28	189.27
14	79.46	172.35
15	78.81	183.69
16	78.31	248.79
17	78.34	404.66
18	64.63	377.05
19	51.01	272.15
20	38.18	198.14
21	29.58	149.00
22	23.36	112.96
23	15.85	75.88
Grand Total	**48.19**	**204.38**



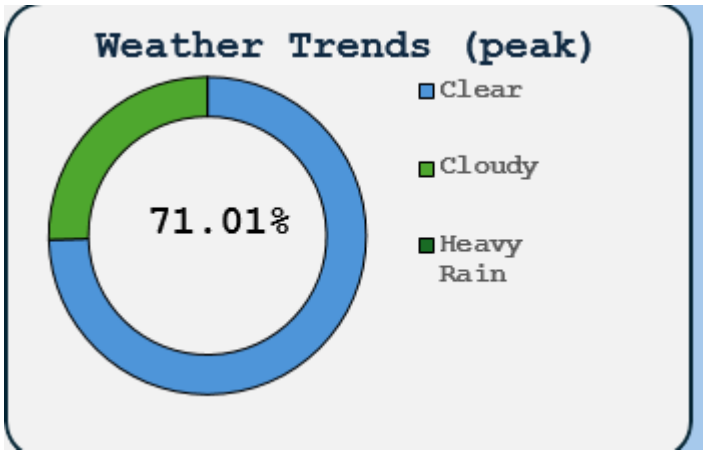
2 Weekly Rental Tendencies

Row Labels	Sum of rentals
0	424350
Sunday	424350
1	435544
Monday	435544
2	441292
Tuesday	441292
3	441428
Wednesday	441428
4	466927
Thursday	466927
5	466290
Friday	466290
6	458517
Saturday	458517
Grand Total	3134348



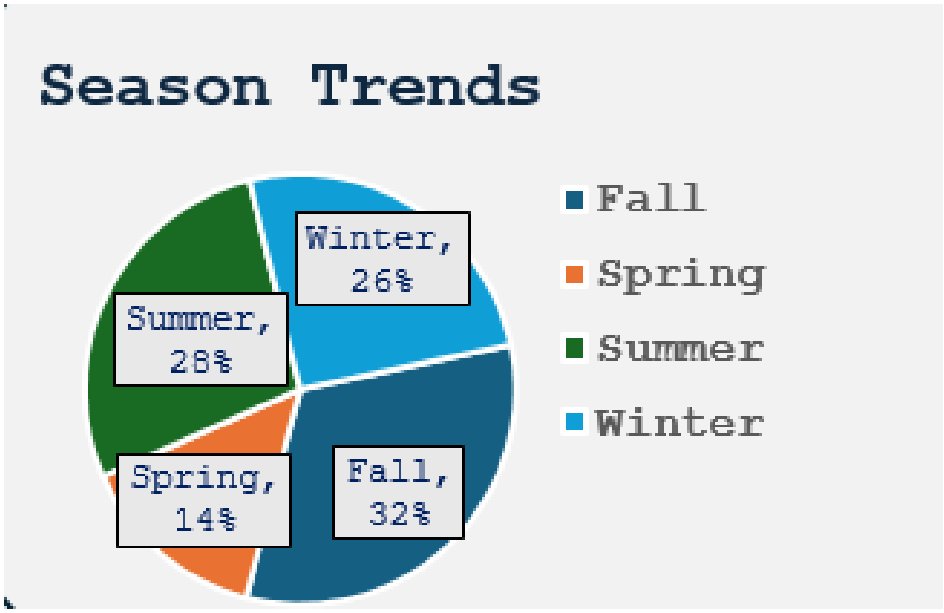
3)Rental distribution by weather

Row Lab	Sum of rentals
Clear	2338173
Cloudy	795952
Heavy Rain	223
Grand Total	**3134348**



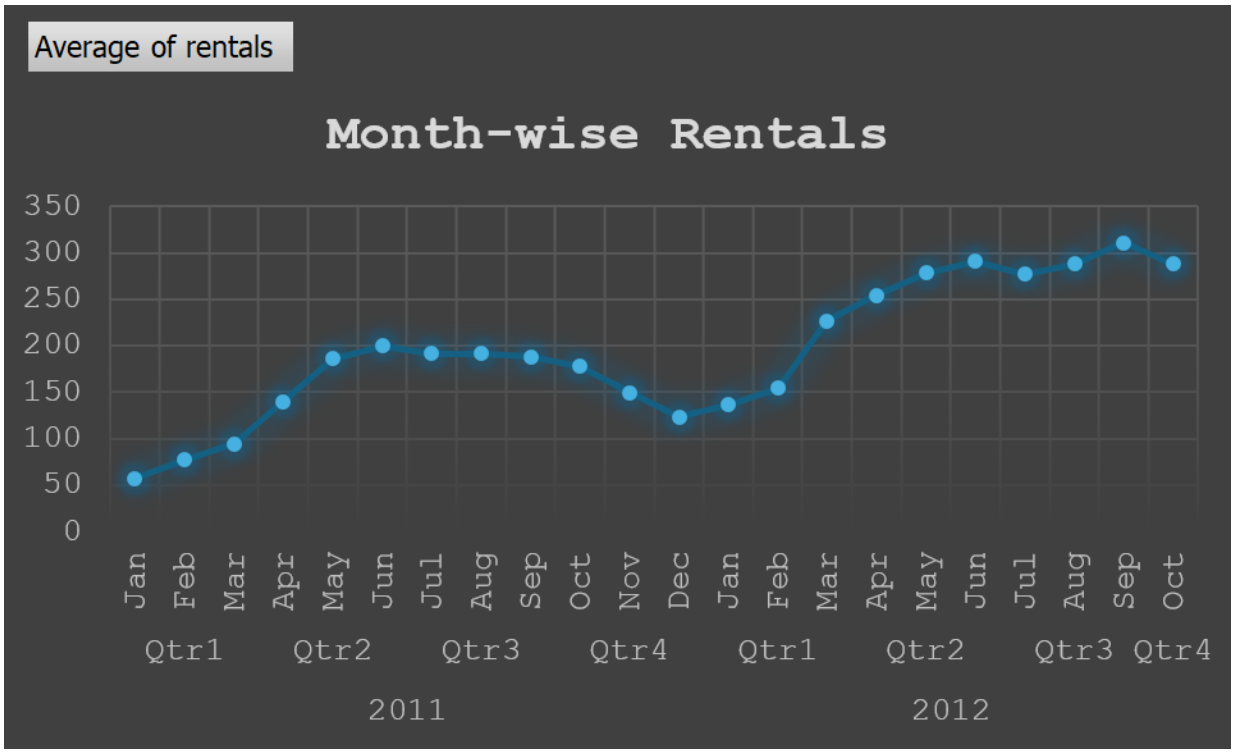
4)Rental distribution by season

Row Lab	Sum of rentals
Fall	1014873
Spring	449706
Summer	871826
Winter	797943
Grand Total	**3134348**



5) Distribution of Renatals monthly

Row Labels	Average of rentals
2011	**149.4923703**
Qtr1	**76.4435527**
Jan	57.26447574
Feb	77.49399657
Mar	94.438949
Qtr2	**176.1906907**
Apr	139.9083728
May	185.9867452
Jun	199.9737609
Qtr3	**190.4125185**
Jul	191.3058985
Aug	191.4304539
Sep	188.2301459
Qtr4	**149.5726626**
Oct	177.4906542
Nov	149.3612403
Dec	123.4537445
2012	**251.187138**
Qtr1	**173.369**
Jan	136.3504399
Feb	154.352657
Mar	226.5337159
Qtr2	**274.8243781**
Apr	254.2326284
May	278.7237237
Jun	291.0043988
Qtr3	**291.7516843**
Jul	277.4418938
Aug	287.7531915
Sep	310.6760355
Qtr4	**288.1364341**
Oct	288.1364341
Grand Total	**196.4000137**


Average of rentals
Month-wise Rentals
350
300
250
200
150
100
50
0
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Qtr1
Qtr2
Qtr3
Qtr4
Qtr1
Qtr2
Qtr3
Qtr4
2011
2012